COMP6237 Data Mining

Lecture 10: Semantic Spaces (Finding Features I)

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Lecture slides available here:

http://comp6237.ecs.soton.ac.uk/zh.html

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Semantic Spaces – Problem

	doc	title
0	m1	Human machine interface for ABC computer applications
1	m2	A survey of user opinion of computer system response time
2	m3	The EPS user interface management system
3	m4	System and human system engineering testing of EPS
4	m5	Relation of user perceived response time to error measurement
5	g1	The generation of random, binary, ordered trees
6	g2	The intersection graph of paths in trees
7	g3	Graph minors IV: Widths of trees and well-quasi-ordering
8	g4	Graph minors: A survey

Are these documents about the same topic?

Semantic Spaces – Problem

Semantic Spaces:

- represent word meanings as vectors that keep track of the words distributional history
- focus on semantic similarity
- similarity measured using geometrical methods

e.g. Cosine similarity between PC and Windows = 0.77
Cosine similarity between PC and window = 0.13
In Japanese, A. Utsumi, IEEE SMC 2010

Semantic Spaces – Latent Semantic Analysis

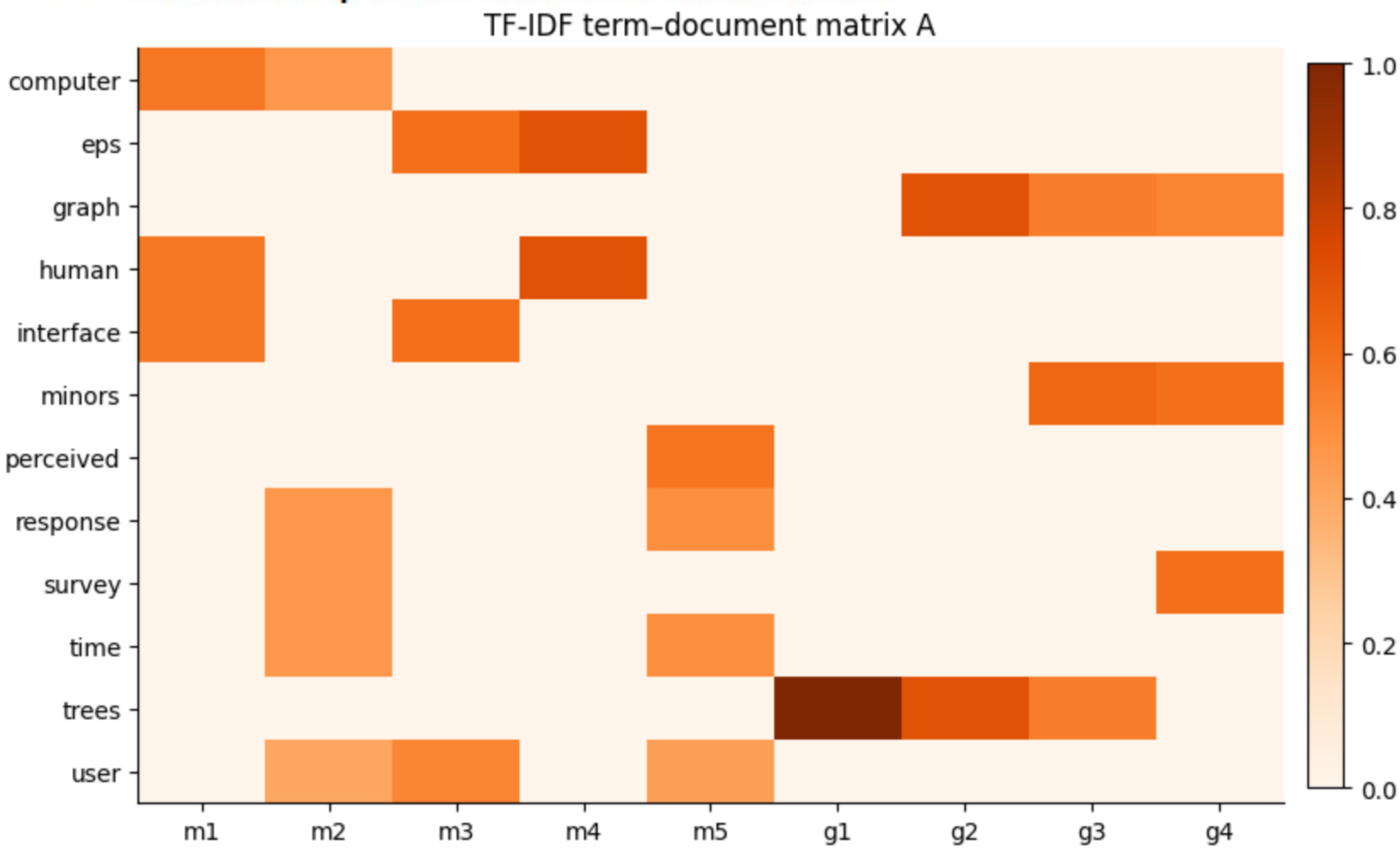
Consider a term-document matrix which describes occurrences of terms in documents Demo: Semantic_Spaces_Visual

	m1	m2	m3	m4	m5	g1	g2	g3	g4
computer	0.58	0.46	0.00	0.00	0.00	0.0	0.00	0.00	0.00
eps	0.00	0.00	0.60	0.71	0.00	0.0	0.00	0.00	0.00
graph	0.00	0.00	0.00	0.00	0.00	0.0	0.71	0.55	0.52
human	0.58	0.00	0.00	0.71	0.00	0.0	0.00	0.00	0.00
interface	0.58	0.00	0.60	0.00	0.00	0.0	0.00	0.00	0.00
minors	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.63	0.60
perceived	0.00	0.00	0.00	0.00	0.58	0.0	0.00	0.00	0.00
response	0.00	0.46	0.00	0.00	0.49	0.0	0.00	0.00	0.00
survey	0.00	0.46	0.00	0.00	0.00	0.0	0.00	0.00	0.60
time	0.00	0.46	0.00	0.00	0.49	0.0	0.00	0.00	0.00
trees	0.00	0.00	0.00	0.00	0.00	1.0	0.71	0.55	0.00
user	0.00	0.40	0.52	0.00	0.43	0.0	0.00	0.00	0.00

Semantic Spaces – Latent Semantic Analysis

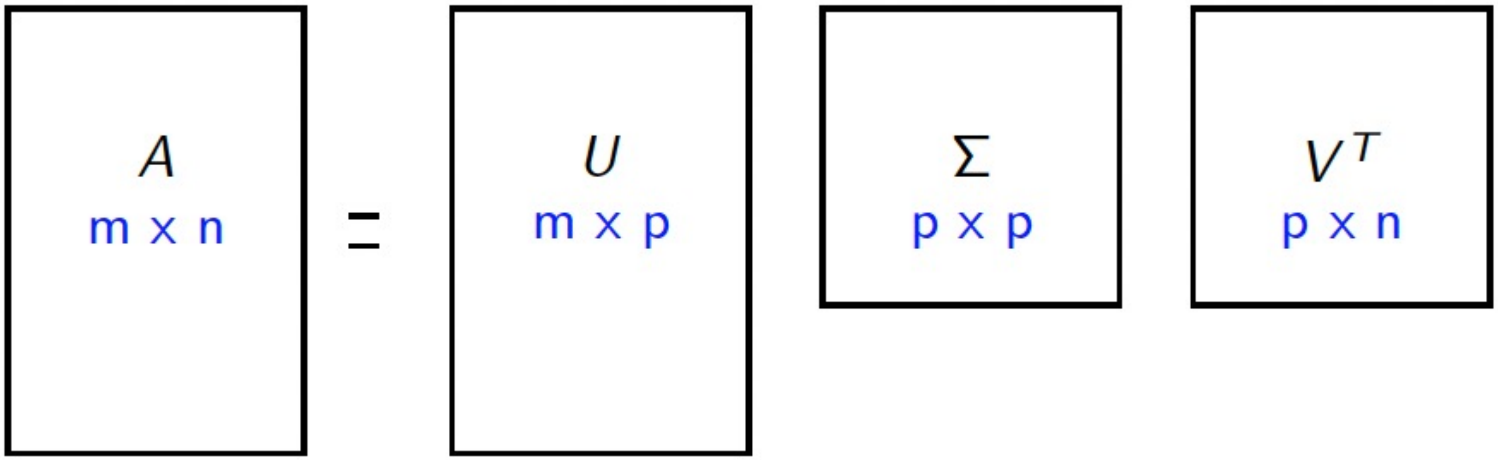
Latent Semantic Analysis (LSA) makes a low-rank approximation
It assumes the term-document matrix:

- is noisy, and should be de-noised
- is more sparse than it should be



Semantic Spaces – Recap SVD

$$A = U \Sigma V^T$$

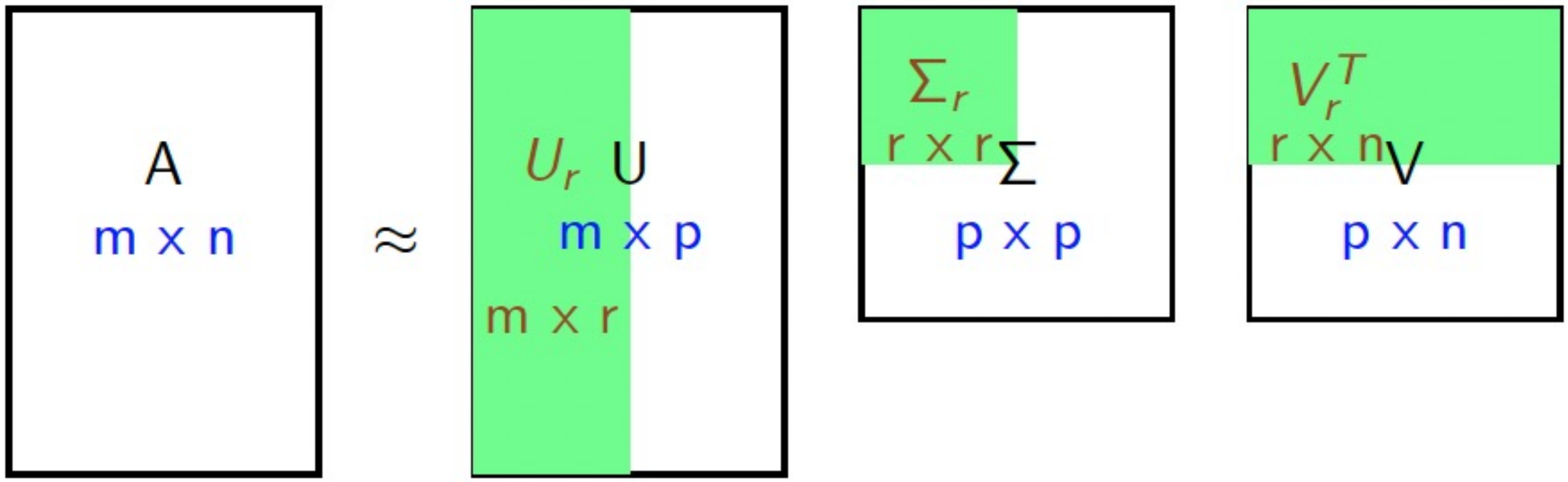


Where p is rank of matrix A

U called *left singular vectors*, contains the eigenvectors of AA^T,
V called *right singular vectors*, contains the eigenvectors of $A^T A$
Σ contains square roots of eigenvalues of AA^T and $A^T A$

If A is matrix of mean centred featurevectors, V contains principal components of the covariance matrix

Semantic Spaces – Recap Truncated SVD

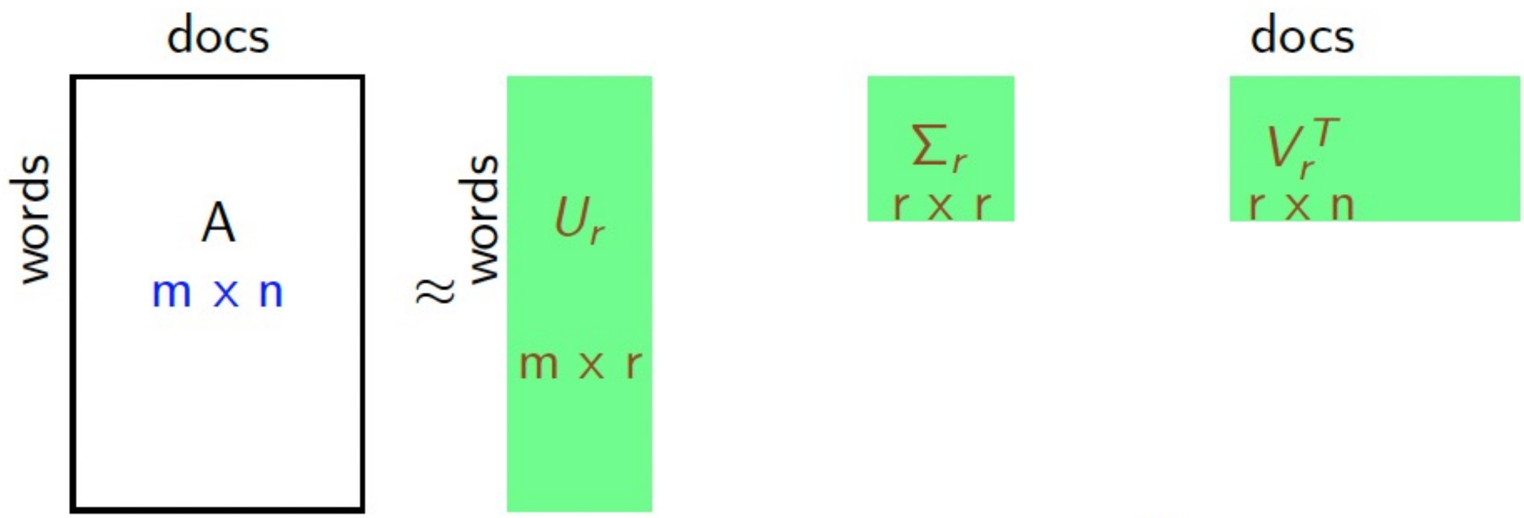


Uses only the largest r singular values (and corresponding left and right vectors)

This can give a *low rank approximation* of A, $\tilde{A} = U_r \Sigma_r V_r$

This has the effect of minimising the Frobenius norm of the difference between A and $\tilde{A}$

Semantic Spaces – Recap Truncated SVD



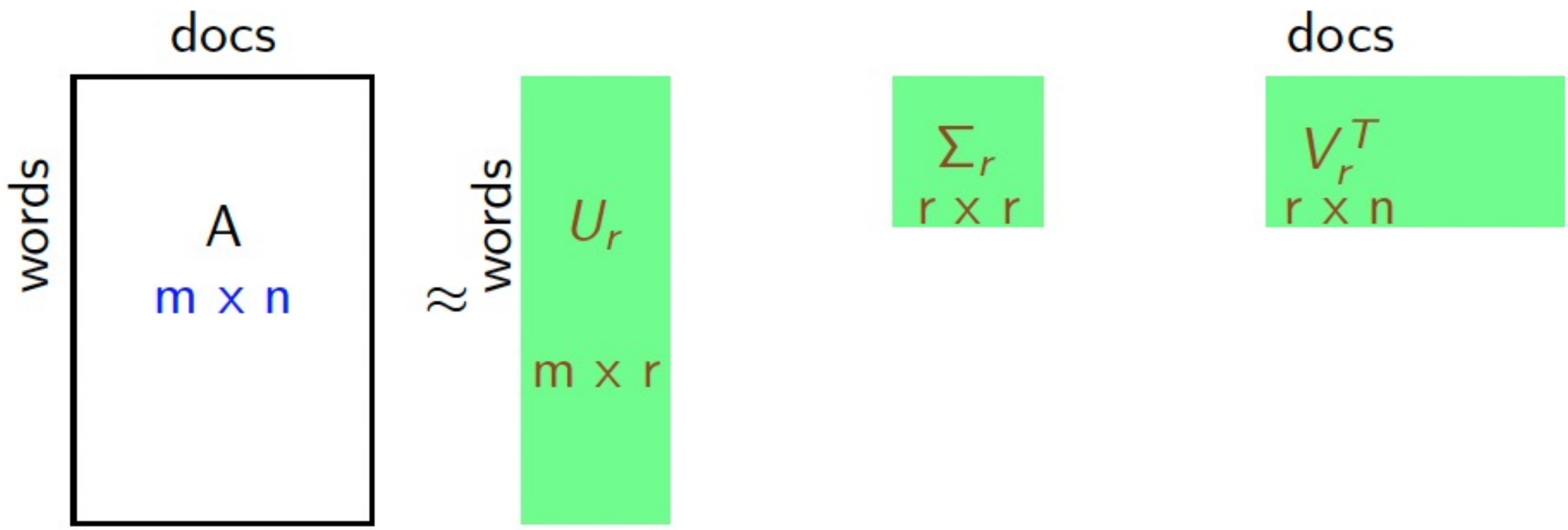
Each row of V_r corresponds to an eigenvector of $A^T A$

- This means it is proportional to the covariance or correlation between the documents
- These are the *concepts*

Each row of U_r describes a term as a vector of weights with respect to r *concepts*

Each column of V_r describes a document as a vector of weights with respect to r concepts

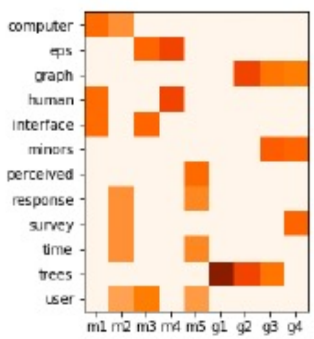
Semantic Spaces – LSA



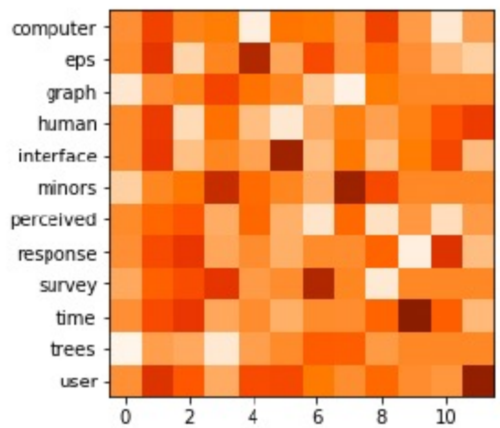
Term concepts and document concepts have the same dimensionality, but represent different spaces.

Semantic Spaces – LSA

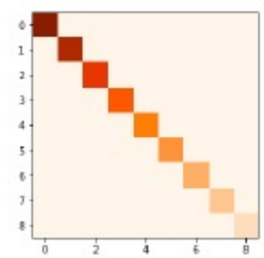
SVD: $A = U \Sigma V^T$



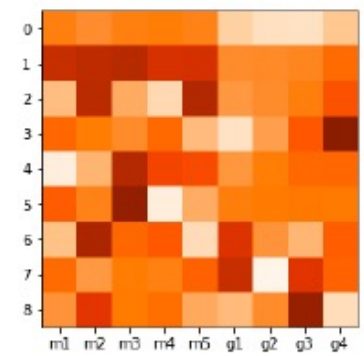
A



U



Σ

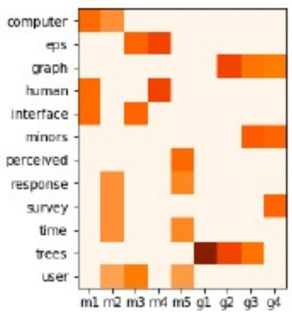


V

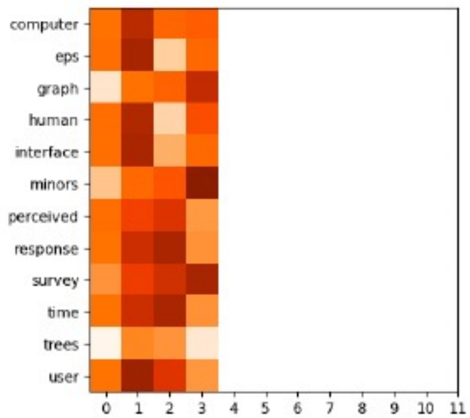
We then reduce the dimensionality by choosing only the first few eigenvalues (in Σ) and the corresponding columns in U and V.

Semantic Spaces – LSA

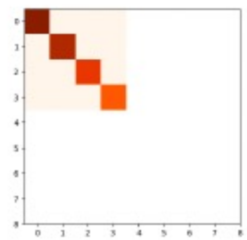
SVD: $A \approx U_r \Sigma_r V_r^T \quad r = 4$



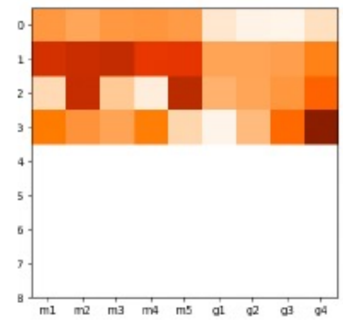
A



U_r



Σ_r

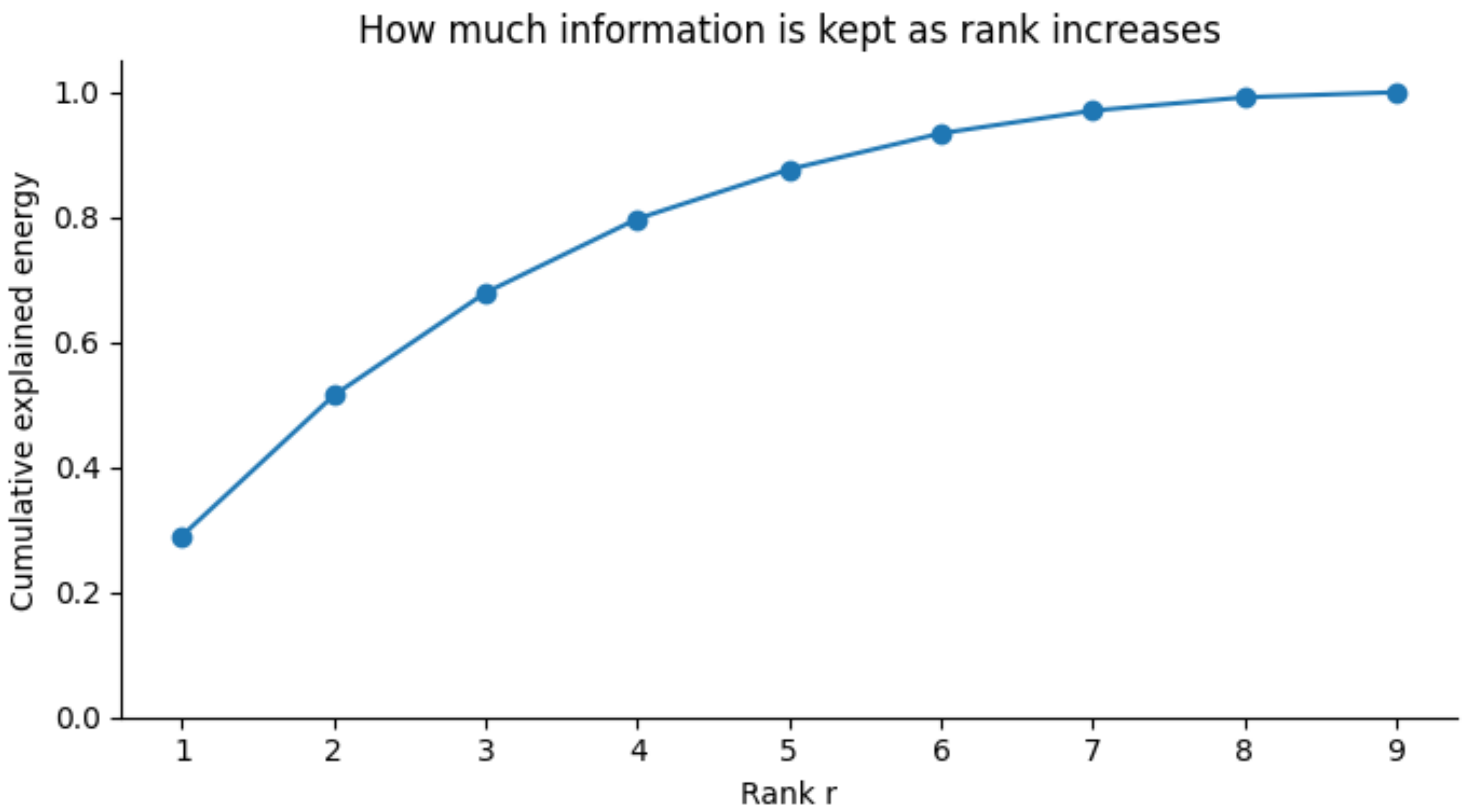


V_r

Each row of U_r describes a word as a vector of weights with respect to r *concepts*

Each column of V_r describes the title as a vector of weights with respect to r concepts

Semantic Spaces – LSA

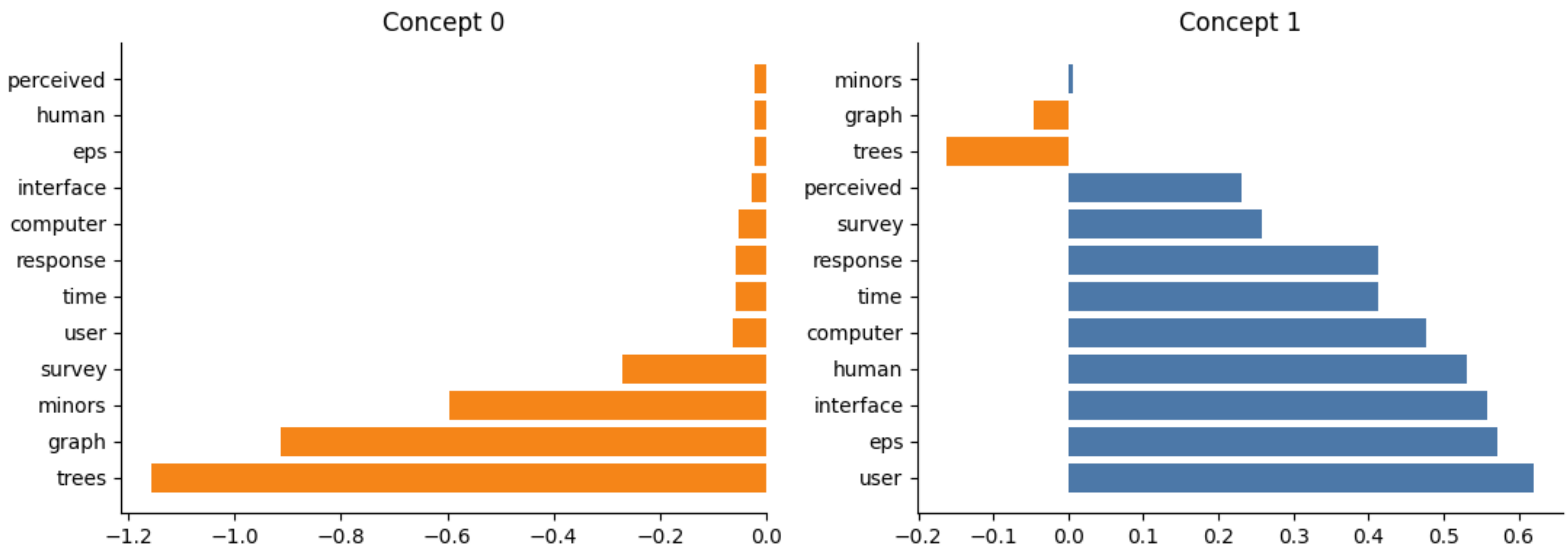


Semantic Spaces – LSA

What do these 'concepts' mean?

U_r gives us the weighting of the words for each concept

We can show that weighting for each concept:



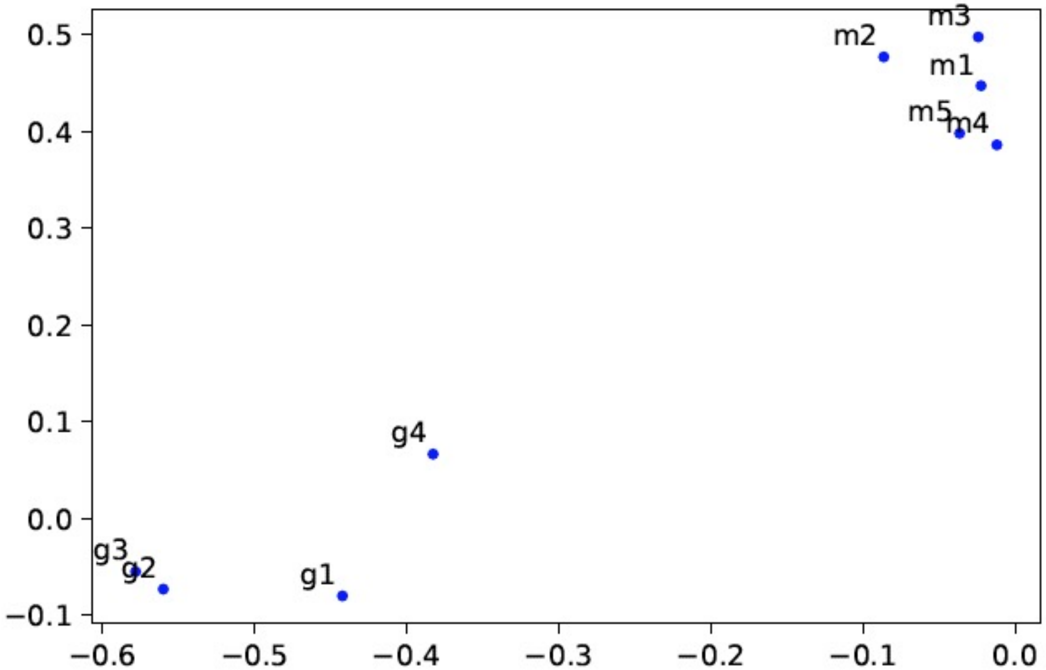
This shows us perhaps that the concepts are not always particularly meaningful.

Demo: Semantic_Spaces_Visual

Semantic Spaces – LSA

Cosine similarity of document vectors can be compared ($r = 2$)

- Vectors for "m1" and "m2" give cosine similarity $= 0.93$
- Vectors for "g1" and "g2" give cosine similarity $= 0.83$
- Vectors for "g1" and "m1" give cosine similarity $= 0.18$




```
r = 2
Ur = U[:, :r]
Sigmar = np.diag(s[:r])
VTr = VT[:r, :]

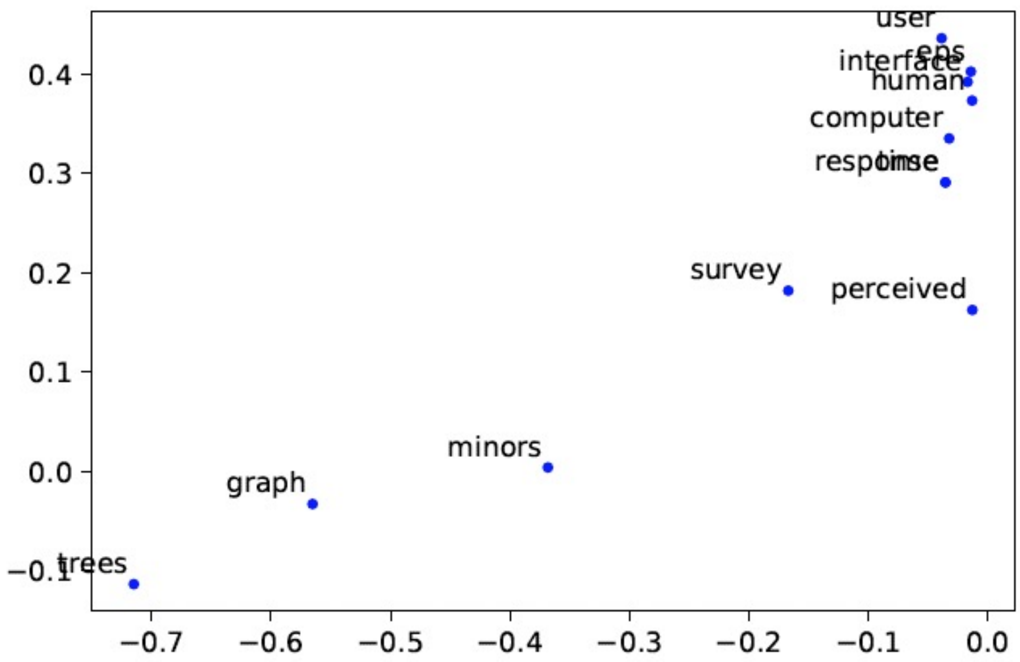
doc_coords = (Sigmar @ VTr).T
word_coords = Ur @ Sigmar
```

Clustering algorithms can be used on the vectors

Semantic Spaces – LSA

Cosine similarity of word vectors can be compared ($r = 2$)

- Vectors for "human" and "interface" give cos similarity $= 0.95$
- Vectors for "human" and "user" give cos similarity $= 0.11$
- Vectors for "graph" and "minor" give cos similarity $= 0.90$

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doc_coords = (Sigmar @ VTr).T
word_coords = Ur @ Sigmar
```

Clustering algorithms can be used on the vectors

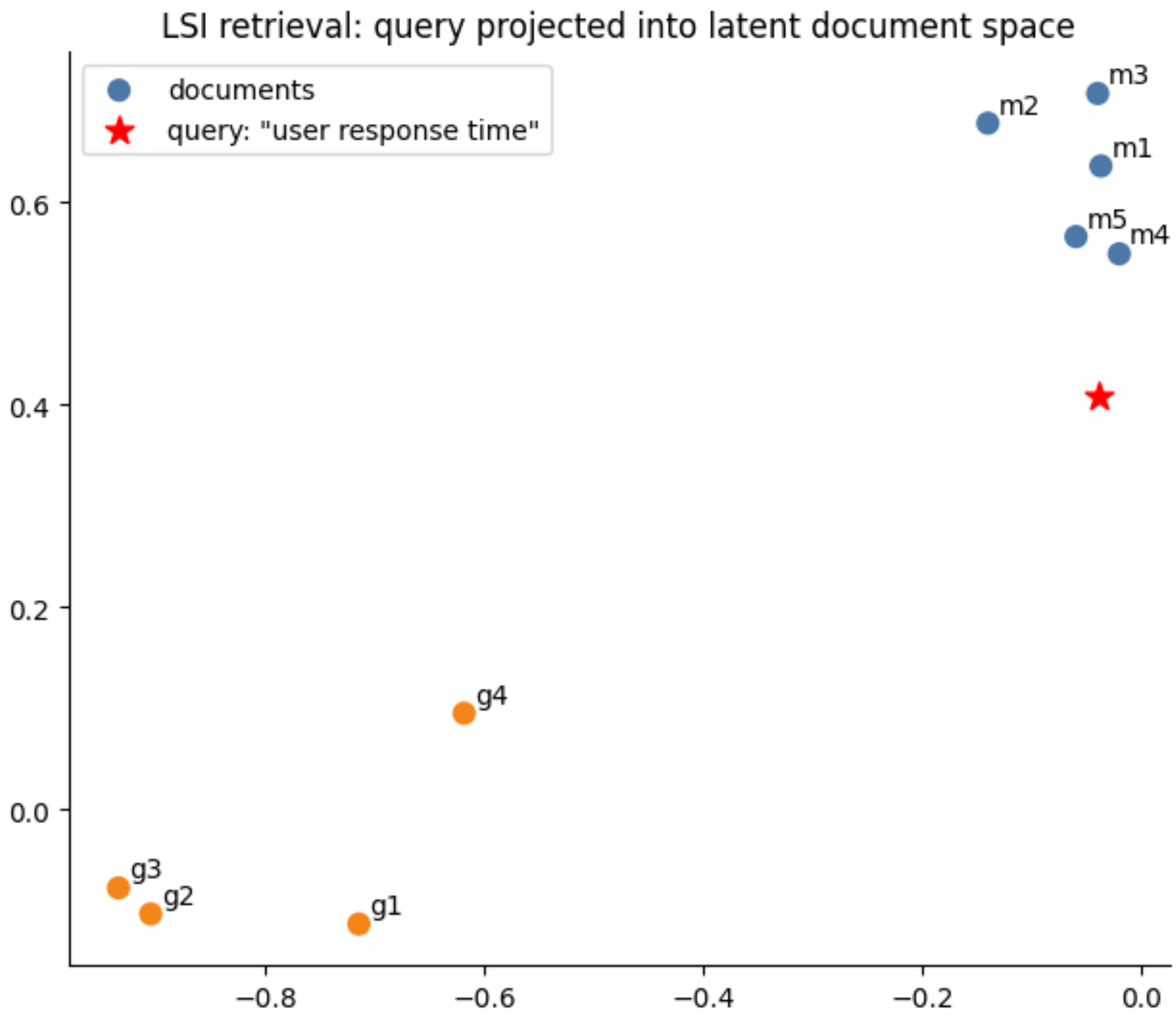
Semantic Spaces – LSI

Latent Semantic Indexing (LSI) LSA can be used for document retrieval

- Given Query: view as query vector $\boldsymbol{q}$
- Project $\boldsymbol{q}$ in to document space
- Compare with document vectors, find closest

Results work mathematically
However, results may not be easy to interpret in terms of natural language.

Semantic Spaces – LSI



Semantic Spaces – LSA

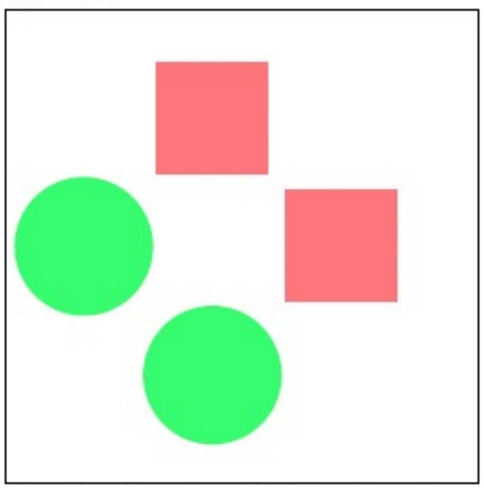
Problems?

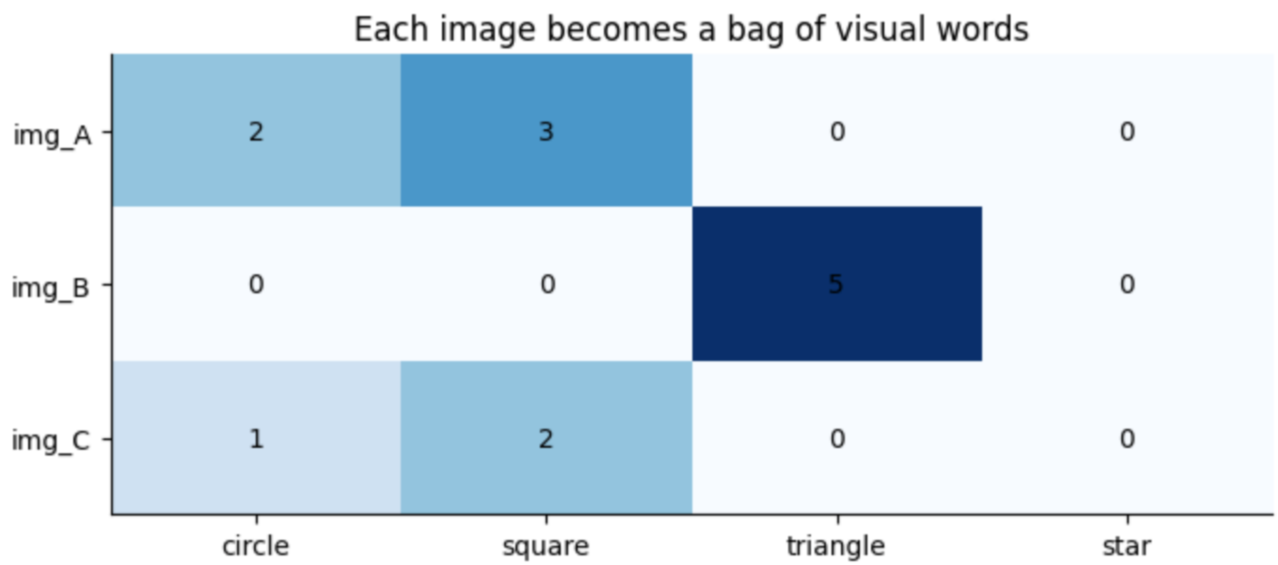
- Polysemious words - with multiple meanings - aren't captured
  - The vector representation averages all meanings of the word
  - e.g. 'fit' is an adjective and a verb
- Word order is ignored (use n-grams?) *An n-gram is a sequence of n words*
- LSA assumes words and documents form a joint Gaussian distribution, however a Poisson distribution is observed

Demo: Semantic_Spaces_Visual

Semantic Spaces – LSA

So far: *bag of words* (BOW) from natural language.
However the maths should work for compositions of occurrences in any unit.
For example, in image search we might want to search for other images with circles.
The image could be encoded with the number of different shapes it has.





BoW:	○	△	□	×	∗
	2	0	3	0	0

Semantic Spaces – LSA

Need to make a large multidimensional space in which images, keywords and visual terms can be placed

In training:

- Learn how images and keywords are related
- Place images and keywords close together in the space

Unannotated images can be placed in the space based on the visual terms they contain

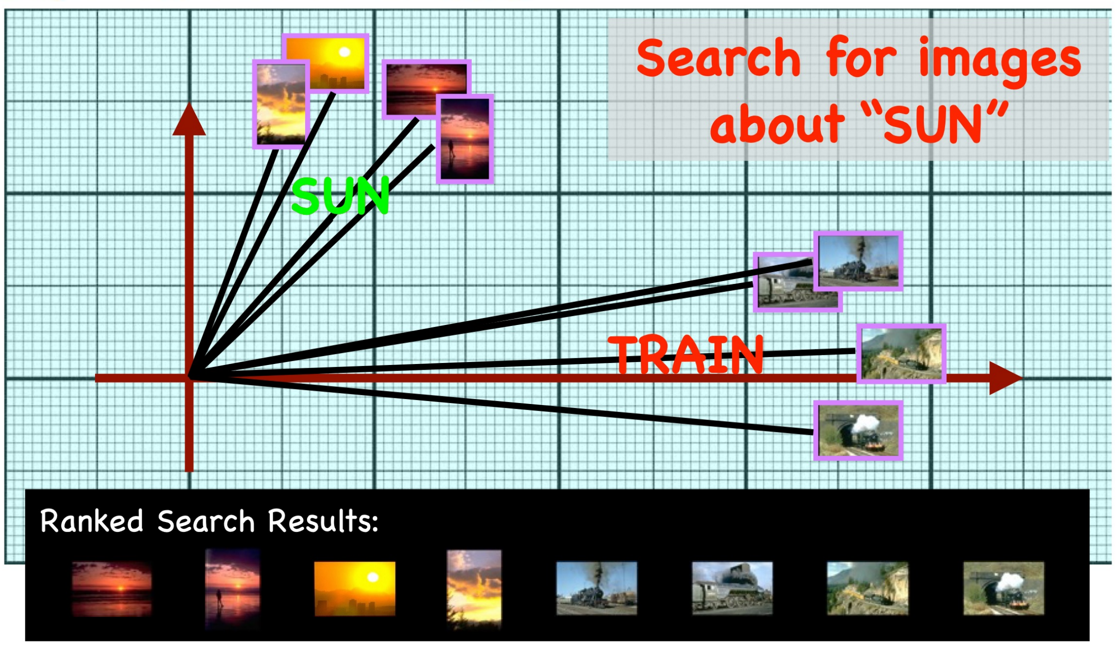
- Images can be placed based on their visual terms in the space
- They should lie near the keywords that describe them

Demo: Semantic_Spaces_Visual

Semantic Spaces – LSA

This lower dimensional space can be used to:

- Find Images using similar words
- Find images with similar images
- Return possible key words for an image
- Find relationships between words, and between words and visual terms



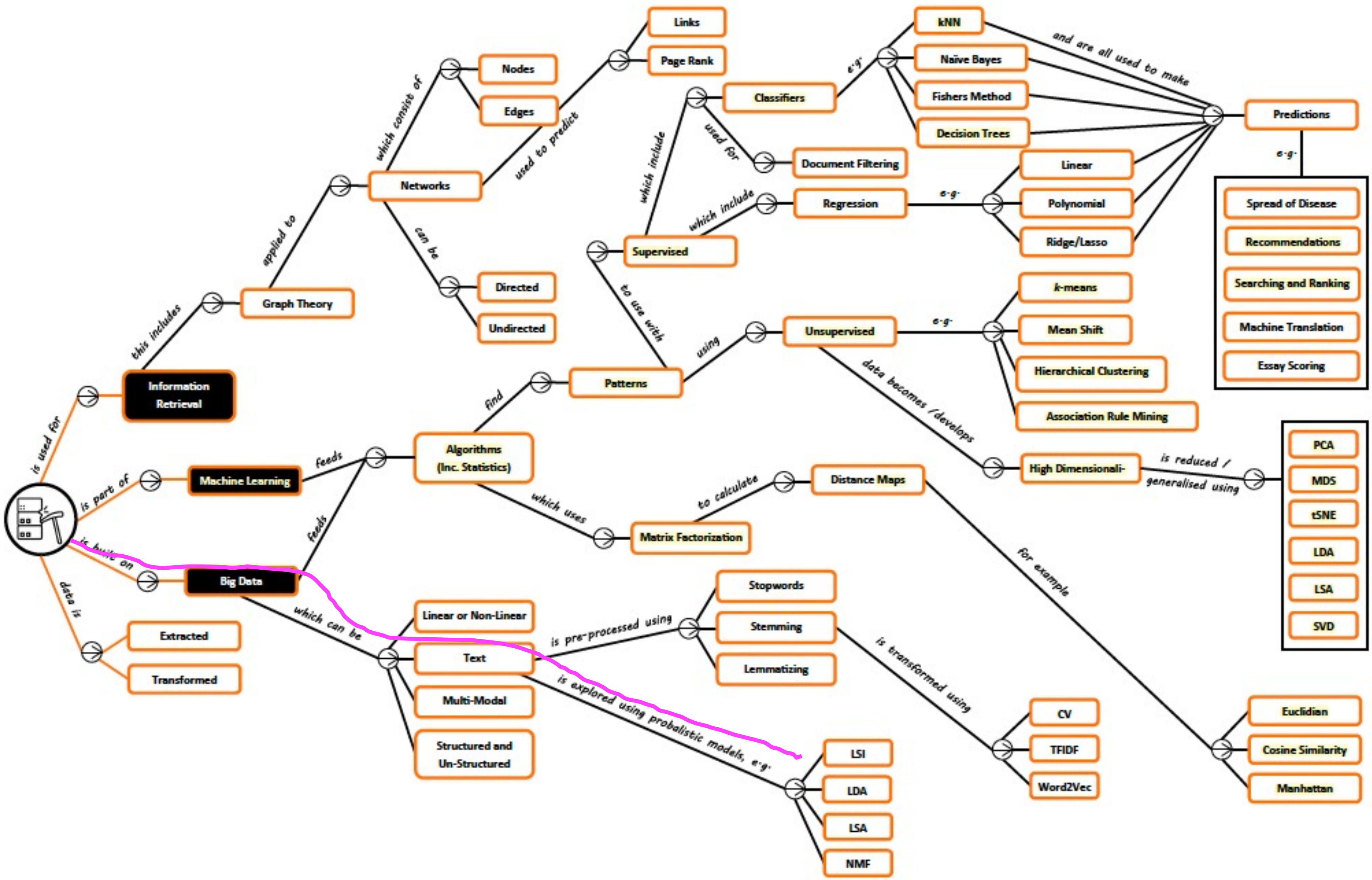
Semantic Spaces – Summary

```
Top query retrieval results for: 'user response time'
```

	doc	title	cosine_with_query
4	m5	Relation of user perceived response time to error measurement	0.999965
0	m1	Human machine interface for ABC computer applications	0.999252
2	m3	The EPS user interface management system	0.999186
3	m4	System and human system engineering testing of EPS	0.998190
1	m2	A survey of user opinion of computer system response time	0.994227

```
Takeaway:
- TF-IDF gives a sparse term-document matrix
- SVD / truncated SVD creates a lower-dimensional latent space
- Cosine similarity in that space captures semantic closeness
- The same idea extends beyond text to visual words and multimodal data
```

Semantic Spaces – **Roadmap**



Finding Features I – **Textbook**

University of Southampton

CHAPTER 10

Finding Independent Features

Most of the chapters so far have focused primarily on *supervised* classifiers, except Chapter 3, which was about *unsupervised* techniques called *clustering*. This chapter will look at ways to extract the important underlying features from sets of data that are not labeled with specific outcomes. Like clustering, these methods do not seek to make predictions as much as they try to characterize the data and tell you interesting things about it.

- Programming Collective Intelligence: Building Smart Web 2.0 Applications *T. Segaran*.

Semantic Spaces – **Learning Outcomes**

- **LO1**: Demonstrate an understanding of techniques for finding independent semantic features, such as: (exam)
  - Comprehending the core concepts of Latent Semantic Analysis (LSA) and apply LSA on a dataset
  - Discussing the advantages and disadvantages of algorithm
- **LO2**: Implement the learned algorithms for independent semantic feature learning (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf*
- *ChatGPT or other AI-based techs*